

Answering causal questions using natural experiments in observational field studies

Abstract

Causation is the knowledge that helps predict the outcomes of specific interventions. Correlational analysis, such as linear regressions, only describes patterns in observed data; it does not imply causations. In this paper, we apply regression discontinuity, a causal inference method that can identify and quantify causations. This method utilizes natural experiment data at thresholds of policies. China's Huai River policy has a geographical threshold for providing cities with district heating in winter. Cities north of the geographical threshold are provided with district heating, while cities in the south are not but might use their own domestic heating. Using the regression discontinuity method, we find that district heating caused indoor operative temperatures to be 4.4 °C higher and the thermal sensation votes of occupants to be 0.6 warmer. Using conventional correlational analysis, we find that the indoor operative temperature is correlated with occupants' thermal satisfaction. However, the correlation could be either positive or negative based on whole-group or subgroup data, respectively. This finding highlights the importance of causal inference methods in thermal comfort field studies, which are mostly observational and usually employ correlational methods.

Keywords

Causal inference; Regression discontinuity; Thermal comfort; Field study; District heating

1. Introduction

Scientists and researchers are often interested in the causal relationship, which is distinct from the correlational relationship. The correlational relationship describes a “seeing” pattern in which a variable changes in unison with another [1]. For example, the number of firefighters at a scene correlates with the amount of damage caused by a fire. However, the firefighters are not the cause of the damage; there are more firefighters because larger fires require a greater response and cause more destruction. Correlational analysis alone can not prove causation and may limit the extent of insights or actionable design or policy recommendations and, in some cases, may even mislead. In contrast, the causal relationship represents a “doing” knowledge that if interventions were made on one variable (the ‘treatment’), it would lead to changes in another (the ‘outcome’) [1–3]. The causal effect is the quantification of the causal relationship, enabling us to better predict an outcome response if an intervention is made in the treatment.

We strive to understand the kind of “doing” causal knowledge and measure the causal effects [4]. This can be achieved by asking and answering causal questions. A causal question asks if a specific intervention could lead to a particular outcome. In building science, we are typically interested in whether interventions in the design, construction, and operation of buildings lead

to improvements in building performance and/or the well-being of building occupants. For example, interventions could include obtaining a green building certification, increasing indoor carbon dioxide concentration, or having a window view. The outcomes could include building energy consumption, occupant satisfaction, work performance, and health. Yet, in many cases, particularly in real-world conditions, there is not a one-to-one relationship between the intervention and the outcome of interest, and there may be many confounding factors and required assumptions. As we will see, this is where data collected from natural experiments is particularly informative.

Research questions in the building science literature are often not framed using specific cause-and-effect language. Many of them indeed ask causal questions even though they may or may not use causal inference methods:

- 1) Do green certification programs save (“*cause*”) energy or improve (“*cause*”) occupants’ satisfaction? [5–12]
- 2) Does a high indoor carbon dioxide concentration affect (“*cause*”) human subjects’ health and work performance? [13–17]
- 3) What is the impact (“*causal effect*”) of a window view on thermal comfort, emotion, and cognitive performance? [18–22]

Some of these causal questions have been answered through the experimental method [13,14,16,21]. Experiments are about “doing” and enable the identification of causal knowledge. For example, to study the causal effect of window views on thermal comfort, researchers created two environmental conditions: one with a window view (treatment group) and another without a window view (control group) [21]. We assume that all other potential factors influencing thermal comfort, such as indoor thermal conditions, occupants’ characteristics, etc., won’t contribute to the outcome if they remain the same between the two groups. Thus, the difference in the outcome between the treatment and groups (e.g., different thermal sensations) can only be attributed to the different experimental conditions. This conclusion would be considered as the causal effect of the window view.

It is not always feasible to conduct an experiment to investigate a particular causal relationship. An observational study might be the only option, especially when an intervention’s outcome might harm human subjects’ health. For example, epidemiologists might not be entirely satisfied with observing the correlation between smoking and cancer. However, it would be unethical to conduct experiments forcing one group of subjects to smoke while another group refrains from smoking to identify the causal relationship to better inform methods for reducing cancer [23–25,1]. Using another building science example, it would be unethical to intentionally reduce ventilation or remove filtration to identify the impact of these interventions on the COVID-19 infection rate in a space. Another common concern of the experimental method is cost and/or time. For example, building science researchers have observed different thermal responses of occupants in air-conditioned and naturally ventilated buildings [26,27]. They might, therefore, consider conducting experiments in two types of buildings. However, such an experiment could be deemed

impractical as it requires a large-scale and long-term control of indoor environments for several months or years to allow for thermal adaptations [28].

It is challenging to perform causal inference in observational studies, which, when expressed another way, would mean understanding “doing” knowledge (causal) from “seeing” data patterns (correlational). In this paper, we draw on our expertise in building science and apply a causal inference framework and method, known as regression discontinuity, to an existing dataset from observational studies of thermal comfort that is particularly suitable for this method. This causal inference framework has been successfully applied to many important research questions in other disciplines [24,29–33]. We compare results using conventional correlational methods versus causal inference methods. The results highlight correlations alone, even with logical reasoning, cannot imply causations.

2. Observational thermal comfort field studies

Thermal comfort research can be categorized into two main methods [34]. The first method is laboratory experiments in climate chambers, where researchers recruit participants, control indoor environmental conditions, and record participants’ thermal responses. The second method is field studies in actual buildings, where researchers either continuously monitor the same building occupant’s responses and corresponding indoor environmental conditions or sample multiple building occupants across one or multiple spaces in a building in a relatively short period. The intention of the field study method is to capture any contextual effects in real-world buildings that may not be fully represented in laboratory experiments or lost due to relatively short experiment durations.

In this paper, we refer to a typical field *study* as an observational study and a field *experiment* as one that involves interventions in actual buildings, such as manipulating or controlling personal or environmental conditions [35,36]. Historically, this distinction has not been consistent, and the terms have sometimes been used interchangeably within the thermal comfort research community. In thermal comfort literature, most field research has been “field studies”, or observational studies that we are more precisely defining here. There are examples of self-described “field studies” that involve randomized and controlled experiments in actual buildings [37,38]. We also find several observational studies that were referred to as “field experiments,” though they don’t include interventions [39–41]. To emphasize the critical difference between the use of experimental data and observational data in causal inference, we will refer to the typical field study as a “field observational study.”

Causal diagrams in Figure 1 help to visualize the critical difference between laboratory experiments (1a), field observational studies (1b), and field experiments (1c). Variables are categorized as causes and effects, and the diagram is drawn based on domain knowledge. For example, we know that indoor temperature affects thermal sensation, so we can draw an arrow pointing from the former to the latter to indicate the causal relationship. In a laboratory experiment (Figure 1a), indoor environmental conditions (e.g., different indoor temperatures), denoted as the treatment node X, are arbitrarily set by the researchers and are not affected by other factors. Thus, there is no arrow pointing to the indoor temperature. Therefore, the

observed relationship τ_1 between the indoor temperature and the outcome, e.g., thermal sensation, denoted as node Y, can be considered as causal effects under the influence of causal factors in the context of a climate chamber. This could be no window views or limited time of exposure, denoted as node E.

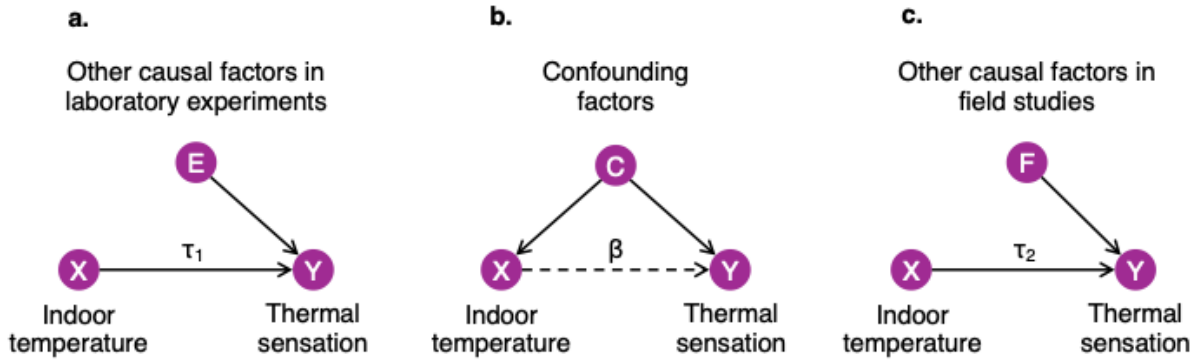


Figure 1 Causal relationships in laboratory experiment (a), typical field study (observational study) (b), and field experiment (c). The arrow represents a directional causal relationship, and the solid line represents the causal effect, denoted as τ . The dashed line represents an observed correlational relationship with the correlational coefficient denoted as β . It has been challenging to measure causal effects in observational studies because of the confounders, denoted as node C. The causal effect τ_1 , measured in a laboratory experiment, is different from τ_2 in a field experiment because of the different influences of other causal factors in the climate chamber and in the actual building, denoted as nodes E and F, respectively.

In a field observational study (Figure 1b), researchers do not interfere with the indoor temperature. Instead, the indoor temperature is typically determined by factors like outdoor weather and climate, types of building heating and cooling systems, occupant preferences, and energy costs. Some of these factors can also affect thermal sensations (e.g., solar radiation affects both indoor temperature and thermal sensations). They are referred to as confounding factors in a causal diagram [42], visualized as node C in Figure 1b, which has solid lines and arrows pointing to both the treatment variable (indoor temperature) and the outcome variable (thermal sensation). Confounders can distort the observed correlational relationship between the treatment and the outcome. Therefore, we visualize the observed relationship β between the indoor temperature and the thermal sensation outcome as a dashed line with an arrow pointing from the former to the latter. This acknowledges the existence of the causal relationship, but the observed relationship does not reflect actual causal effects because of the confounders in a field observational study.

In a field experiment (Figure 1c), researchers manipulate and control the indoor temperature in an actual building, similar to a laboratory experiment. The indoor temperature is not intended to be affected by other factors in the field. Therefore, ideally, there would be no confounders in this setup. The observed relationship τ_2 between indoor temperature and thermal sensation can be interpreted as the causal effect under the influence of other causal factors in the context of an actual building. These could be the outdoor climate, window views, or the types of building heating and cooling systems, denoted as node F, which is different than node E.

3. Causal inference framework

3.1. Defining causal effects from randomized experiments

The causal effect is defined as the difference between potential outcomes of the same unit, which could be either a single person or a group of people [43,44,2]. The potential outcome is a counterfactual concept, which refers to what would have happened to the unit if the treatment had been different in an experiment. For example, if the treatment is bedroom ventilation and the unit is a person, then two potential outcomes could be: the sleep quality of that person after the night sleeping in a bedroom with a high ventilation rate versus the same person's sleep quality after the night sleeping in the same bedroom but with a low ventilation rate. The fundamental challenge of causal inference is that we cannot observe both potential outcomes for a single unit simultaneously. If we expose a person to a well-ventilated bedroom and observe good sleep quality the following day, then we would never know whether the same person would report good sleep if they had been exposed to a poorly ventilated bedroom that same night. For unit i , the causal effect is the difference between the two potential outcomes at the same time. The potential outcome with the treatment is Y_1 , and the potential outcome without the treatment is Y_0 . Then, the causal effect for the unit i is:

$$\tau = Y_1 - Y_0 \quad (1)$$

It is impossible to calculate the causal effect at the individual level. Even if a treatment is applied at two distinct times, the person will not be the same. However, we could estimate the causal effect by assuming the person remains the same during a short period of time or at a population level by considering the two groups are the same, one under the treatment condition and another one under the control condition. The central limit theorem states that the sample mean will converge to a standard normal distribution as long as the same size is large enough. Therefore, the average building occupant of the treatment group and of the control group could have the same characteristics, such as average age, weight, etc. We can then calculate the difference between two average potential outcomes in the two groups as the estimation of the average causal effects. This is also why the causal effect estimated at the population level may not work on individuals. For example, smoking causes cancer in a general sense, but each smoker may not necessarily get cancer. In other words, the causal effect has a probability. Exceptions could occur but have a low probability. The average causal effect estimated at the population level is defined as:

$$\hat{\tau} = \frac{1}{N_1} \sum_{i=1}^{N_1} Y_1 - \frac{1}{N_0} \sum_{i=1}^{N_0} Y_0 \quad (2)$$

The 'gold standard' of causal inference is a randomized and controlled experiment [44–46] where all subject units are randomly assigned to either the treatment or the control group. Therefore, as long as the number of subject units is large enough, the randomization process will eventually cancel out differences between the two groups except in the treatment. We can consider two groups as identical, and the differences observed in outcomes between the two groups can be attributed to the treatment, often referred to as the average treatment effect or causal effect [47]. Two potential outcomes can be considered as two random variables with

the same probability (due to the random assignment), and we can represent the average causal effect as the difference between the expectations of the two random variables.

$$\hat{\tau} = \mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] \quad (3)$$

Another advantage of the random assignment is that it can cancel out confounding factors that we cannot measure but may influence the outcome. In contrast, in observational studies without random assignment, we might be able to find two very similar groups, but there are always other unmeasured or even unknown factors that could confound and limit our capability to identify a causal relationship. For example, we might find two groups of people with similar lifestyles, such as the amount of alcohol use, physical exercise, etc. One group of people smokes while the other doesn't, and we can conduct the data analysis within different age groups, as age is often measured and is the confounder that influences both smoking behavior and cancer. However, we couldn't measure the genetics of the two groups. One could argue that the higher rate of cancer in the treatment smoking groups could be due to special genetics, which also makes people addicted to smoking [23]. Such arguments would be invalid if random assignment were allowed, as the two groups would have a similar distribution of genetics.

3.2. Generalizing causal effects to observational studies

In a typical observational thermal comfort field study, researchers measure many potential confounding factors, such as the building characteristics (e.g., office versus school, building age, orientation, etc.) or personal variables (e.g., demographics, job titles, or workplace location, etc). They might conduct analysis on subsets of the dataset to limit some of the variability within any of those confounding factors. The intention is to control the confounding effect to better understand the relationship of interest. For example, in adaptive thermal comfort field studies, the building cooling type (air-conditioned or naturally ventilated) can affect both the indoor temperature and the occupants' behavior, expectations, and thermal responses in a variety of ways. By setting the building cooling type as the unit of analysis, we can stratify and eliminate the confounding effects. However, there are many other confounding factors, including those we cannot measure or are yet unknown, that could have significant impacts on the outcome. This problem cannot be solved through a large sample size and remains the main challenge of causal inference in observational studies.

The opportunity for causal inference in observational studies arises when random assignment happens in a natural way, which is also referred to as the natural experiment or quasi-experiment [44,48]. In a natural experiment, the difference between the treatment and control group can be solely attributed to different treatment effects. One natural experiment method is regression discontinuity [49–51], which is based on the idea that some rules or policies in the real world are like treatments in the experiment, and the chance that people get the treatment at the threshold of the policy implementation can be considered random [52].

One classic example of applying the regression discontinuity method is estimating the causal effect of alcohol consumption on mortality [30,53]. While there is an observed correlation

between alcohol consumption and traffic incidents [54], the correlation alone does not imply causation, nor can the correlation coefficient be interpreted as the causal effect. For example, individuals who consume more alcohol might do so due to long-term mental health issues that contribute to more traffic incidents. Epidemiologists and economists aim to determine the causal effect of alcohol consumption on traffic incidents, which is critical for public safety-related policymaking. The legal drinking age in the U.S. is 21, which is a policy that provides an opportunity for regression discontinuity research. Whether a young adult's age is just below or just above the 21 threshold is random at the specific date when the data was collected. The drinking policy is like a natural experiment, which randomly assigns those just above age 21 to the treatment group with more alcohol consumption and those just below age 21 to the control group with less alcohol consumption. We can, therefore, consider the difference in mortality near the threshold between the two groups as the causal effect of increased alcohol consumption [30].

3.3. Measuring causal effects at a discontinued threshold

To measure the causal effect, we first need to define a policy discontinuity variable that has the value of 0 at the threshold, called the running variable. In the example of the legal drinking age, the running variable is calculated by a person's actual age minus 21. The equation of causal effect is represented as (4), within which the expectation of the potential outcome $\mathbb{E}[Y_{1i}]$ can be estimated based on a regression model using the data where the running variable is larger than 0, shown in (5). $\mathbb{E}[Y_{1i}]$ would equal the intercept α_1 when the $x_i = 0$. The expectation of the potential outcome $\mathbb{E}[Y_{0i}]$ can be estimated by another regression model using the data where the running variable is less than 0, shown in (6), and is equal to the intercept α_0 when the $x_i = 0$. Therefore, the causal effect, which is the difference between two potential outcomes, equals $\alpha_1 - \alpha_0$ at the threshold of the running variable, shown in (7) and is visualized as the intercept jumps at the threshold in Figure 2.

$$\hat{\tau} = \mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}], \quad x_i = 0 \quad (4)$$

$$y_i = \alpha_1 + \beta_1 x_i + \varepsilon_{1i}, \quad x_i \geq 0 \quad (5)$$

$$y_i = \alpha_0 + \beta_0 x_i + \varepsilon_{0i}, \quad x_i < 0 \quad (6)$$

$$\hat{\tau} = \alpha_1 - \alpha_0, \quad x_i = 0 \quad (7)$$

It is convenient to combine the two regression models, (5) and (6), so that we can utilize existing regression software tools and quickly derive statistical summaries, such as the *p-value* and 95% Confidence Intervals (CI). To achieve this, we need to define a dummy variable D_i that has the value of 1 or 0, depending on whether the running variable is larger than or less than 0, as shown in equation (8). The resulting reduced form of the regression model is (9), and the causal effect $\alpha_1 - \alpha_0$ is represented as the model coefficient of the dummy variable D_i .

$$D_i = \begin{cases} 1, & x_i \geq 0 \\ 0, & x_i < 0 \end{cases} \quad (8)$$

$$y_i = \alpha_0 + (\alpha_1 - \alpha_0)D_i + \beta_0 x_i + (\beta_1 - \beta_0)D_i x_i + \varepsilon_i \quad (9)$$

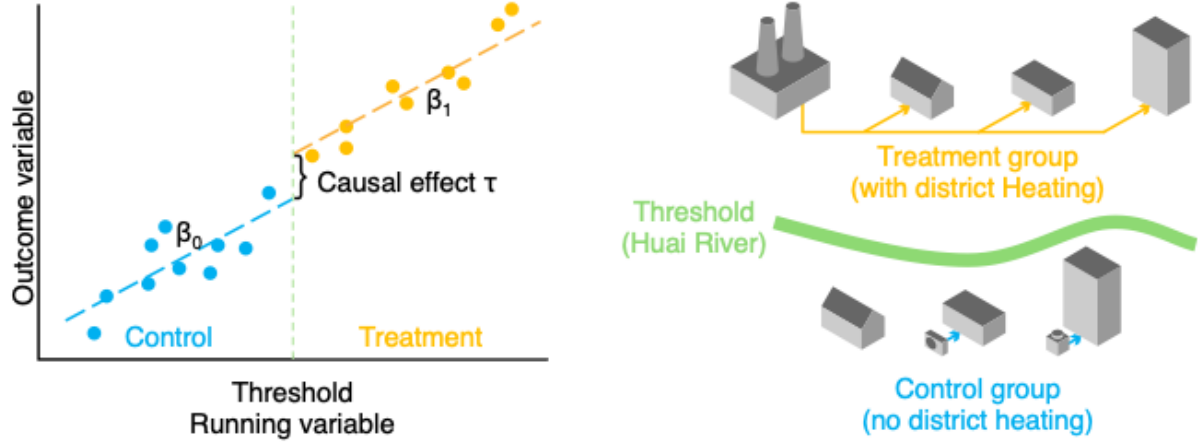


Figure 2 The discontinuity of the winter district heating policy at China's Huai River provides a natural experiment opportunity, enabling causal inference in the observational thermal comfort field study. We define the relative latitude degree to Huai River as the running variable. The difference between the treatment group and the control group at the threshold is considered the treatment effect, often referred to as the causal effect.

4. Case study

There is an opportunity to apply the regression discontinuity method to thermal comfort field studies in China because of an arbitrary boundary (Qingling-Huaihe Line, referred to as the Huai River) of the district heating policy, which is often referred to as the Huai River Policy [55]. The policy provides free or highly subsidized winter heating to offices and homes in cities via central plants burning coal. However, the policy only extends to the north of the Huai River due to budget constraints [56]. The boundary line generally flows along the latitude 33 °N and the 0 °C isotherm in January, historically separating Northern and Southern China. Cities in the south are not provided with district heating but may use their own domestic systems, such as electric heaters and heat pumps. The buildings in the furthest south cities typically have no heating due to relatively warm outdoor temperatures in winter. The geographical boundary of the policy can be considered as the threshold in regression discontinuity research design, and the running variable is the relative distance to the threshold [57].

China's Huai River policy has been previously utilized in multiple research applications of regression discontinuity. For example, researchers estimated the causal effect of district heating on building energy consumption [58]. With a similar regression discontinuity research design, we can measure the causal effects of the district heating on several different outcomes of interest by replacing the y-axis variable in Figure 2. We use the Chinese thermal comfort dataset [59], which contains observational data from thermal comfort field studies across China. There are 15,992 records collected in both Northern and Southern cities during the winter season. Each individual record has simultaneous measurements of the physical environment (e.g., operative temperature) and thermal perception (e.g., thermal sensation

vote). Interested readers can refer to Yang et al., 2023 [59] for a detailed dataset description and summaries of the constituent studies.

We grouped the records by city and added the latitude, resulting in a total of 33 cities, with 14 in the south and 19 in the north. We also manually corrected several data points. Importantly, all data points in Baotou (a northern city) were incorrectly labeled as naturally ventilated buildings rather than radiator heating, which is a common terminal form of district heating. This error has been confirmed by the original data contributor [60]. The cleaned Chinese thermal comfort dataset and the code to reproduce the analysis are available in a GitHub repository that is attached at the end of this paper.

5. Causal effects estimation

To demonstrate causal effect estimations, we used two commonly available variables from the Chinese thermal comfort database as the outcomes of interest, indoor operative temperature, and thermal sensation vote. Thermal sensation votes were collected using a seven-point scale ranging from -3 (cold) and 0 (feeling neutral) to +3 (hot). We calculated the mean values of these two variables for all building types in each city, which include offices, homes, and classrooms. Building type is often considered a confounding factor in most thermal comfort field studies. The benefit of a natural experiment such as this one is that all confounding effects, including those measured or unmeasured, can be canceled out in theory. Therefore, the estimated results represent the causal effects on an average indoor thermal environment during the heating season in a city and on the thermal response of an average occupant.

We use the linear model in equation (9) to introduce a causal effect for simplicity. However, the linear model might have the problem of misattributing the nonlinearity at the threshold as the discontinuity and interpreting it as the causal effect. This issue can be solved using quadratic models. The representation of the causal effect remains the same as linear models, which is still the difference between the intercepts of two regression models below and above the threshold or the model coefficient of the dummy variable D_i in equation (10). We found that linear and quadratic models have similar estimation results of causal effects. We chose quadratic models to display in figures because they have smaller p-values and 95% CI, as shown in Table 1.

$$y_i = \alpha_0 + (\alpha_1 - \alpha_0)D_i + \beta_0x_i + (\beta_1 - \beta_0)D_ix_i + \gamma_0x_i^2 + (\gamma_1 - \gamma_0)D_ix_i^2 + \varepsilon_i \quad (10)$$

Table 1. Estimated causal effects of district heating on indoor temperature and thermal sensation using linear and quadratic models

Outcome	Model	Causal effect	p-value	95% CI
Indoor operative temperature (°C)	Linear	4.9	0.022	0.7 and 9.0
	Quadratic	4.4	0.005	1.4 and 7.4
Thermal sensation vote	Linear	0.6	0.020	0.1 and 1.2
	Quadratic	0.6	0.002	0.3 and 1.0

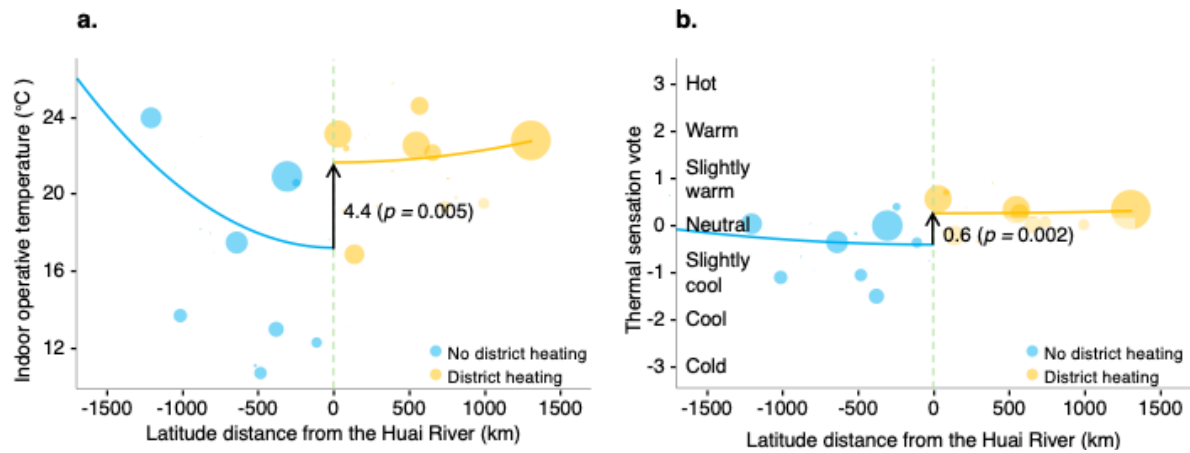


Figure 3 Causal effects of district heating on the mean indoor operative temperature (a) and on mean thermal sensation vote (b). Each data point represents a city, and regressions are weighted based on sample size, represented as the area of each data point. The causal effect is represented as the intercept jump between regressions. The district heating increases indoor operative temperature by 4.4 °C and thermal sensation vote by 0.6. Both causal effects are statistically significant.

The slopes of regression models above and below the threshold in Figure 3 only indicate correlations, and we can only hypothesize causations that lead to the correlations. For example, indoor temperatures get cooler in southern cities as they get closer to the Huai River. This might be due to the cooler outdoor climate as the north latitude degree increases. Above the threshold, operative temperatures inside buildings provided with district heating are relatively flat regardless of outdoor climate. This might be due to the design objective of district heating networks to maintain indoor temperatures at constant levels, which can be achieved by controlling the amount of heating delivered to cities based on the outdoor temperature.

At the threshold of the policy at the Huai River, we can identify the causation and quantify the causal effects. District heating increases the mean indoor operative temperature by 4.4 °C ($p = 0.005$), from 17.1 °C to 21.5 °C. It also increases the mean thermal sensation vote by 0.6 ($p = 0.002$), from -0.4 (cooler than neutral) to 0.2 (warmer than neutral). We further infer that the 4.4 °C increase in indoor operative temperature causes the 0.6 increase in the thermal sensation vote if we could assume that the district heating policy mainly affects building occupants' thermal sensation votes through the indoor operative temperature. Other parameters, like relative humidity and clothing insulation, can also affect the thermal sensation vote, but here, we consider them as consequences of indoor operative temperature changes rather than directly affected by the district heating.

We compared the estimated causal effect of 4.4 °C indoor operative temperature change with prediction results using the Predicted Mean Vote (PMV) model, which is considered a causal model developed from laboratory experiments [3]. We calculated PMV at the threshold for no district heating (control group) and for district heating (treatment group) using the CBE thermal comfort tool based on the ISO standard [61]. We used the same method of estimating the causal effects on indoor operative temperature to calculate other input parameters of the PMV model. For example, the district heating decreases the mean indoor relative humidity level by 25% ($p < 0.001$), from 56% (control group) to 31% (treatment group). We didn't find

causal effects of district heating on mean clothing insulation, air velocity, or metabolic rate. The mean air speed and metabolic rate in the two groups are the same: 0.1 m/s (still air) and 1 met (seated quietly), and differences of these parameters between the two groups are not significant: 0.03 m/s ($p = 0.240$) and 0.01 met ($p = 0.704$). Though the mean clothing insulation is 1.1 clo in the control group and 0.9 clo in the treatment group, the difference is not significant ($p = 0.071$). As a result, the calculated PMV for the control group is -1.39, the calculated PMV for the treatment group is -0.79, and the delta PMV is 0.6, which matches the causal effect estimation.

6. The correlation paradox

The conventional data analysis approach in many observational thermal comfort field studies is correlational analysis. We, therefore, applied a typical correlational analysis, linear regression, to the Chinese thermal comfort dataset to show the limitations of this approach. We first calculated the thermal satisfaction rate for each city as the percentage of individual thermal sensation votes between ± 2 (excluded), which is considered equivalent to thermal satisfaction if the scale resolution is limited to integers, according to the ASHRAE-55 standard [62]. Using the conventional linear regression, we found a positive correlation ($p = 0.016$, $R^2 = 0.04$) between indoor operative temperature and thermal satisfaction rate, as shown in Figure 4a. However, the correlation *does not necessarily imply* a causal relationship. In Figure 4a, within the measured range, higher indoor temperatures are associated with higher thermal satisfaction, but we *cannot infer, based only on this correlation*, that increasing indoor temperature can increase thermal satisfaction. We can hypothesize that the correlation is due to the causation using domain knowledge, for example, obtained in previous experiments. We could then state that people are often happier with a warmer indoor temperature in contrast with the cold outdoor weather in winter; this response is thermal alliesthesia [63].

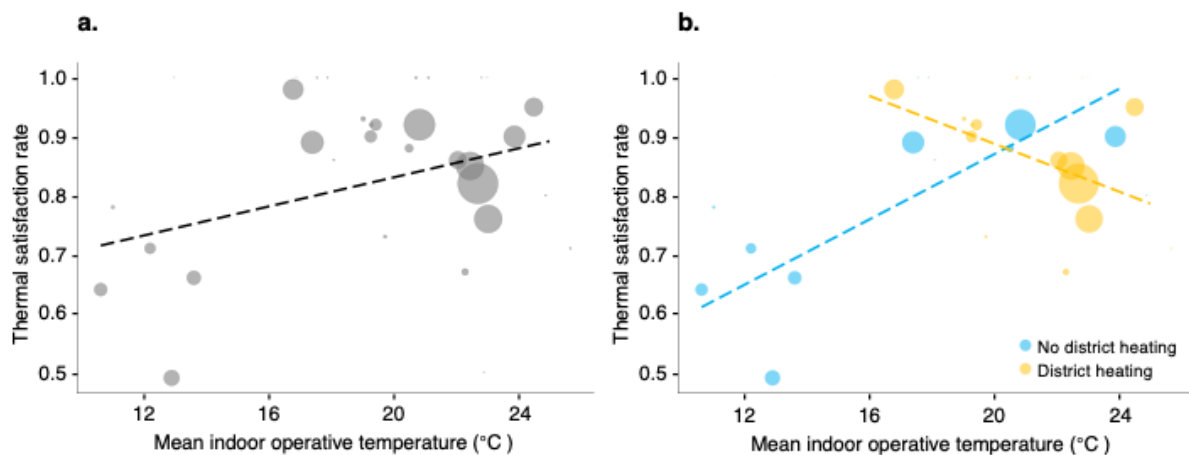


Figure 4 The paradox of correlations between indoor operative temperature and thermal satisfaction rate. The linear regression on the whole dataset shows a positive correlation, while the linear regression in the no district heating subgroup shows a negative correlation.

We divided the data into two groups (one with district heating and another without) and conducted the conventional linear regression for each group. Manipulating the data in this

way is equivalent to conducting field studies and collecting observational data from buildings that have or do not have district heating. As shown in Figure 4b, correlations in the two groups display the opposite trend. The district heating group displays a trend that is opposite to the one in the no district heating group as well as the one in the whole group in Figure 4a. The regression model for the no district heating group is $y = 0.03x + 0.32$ ($p < 0.001$, $R^2 = 0.09$). In contrast, it is $y = -0.02x + 1.29$ ($p = 0.011$, $R^2 = 0.16$) for the district heating group; higher indoor operative temperatures are associated with lower thermal satisfaction rates in buildings with district heating, but they are associated with higher thermal satisfaction rates in buildings with no district heating.

The seemingly contradictory result between Figure 4a and 4b is known as Simpson's paradox [64], which occurs when certain variables are controlled or uncontrolled. Whether we should control a variable depends on the causal relationships between those variables, the treatment, and the outcome of interest. In our case study, if we had explored all potential causal relationships and identified that the building operation mode is a confounding variable, then we should control for it and use the correlations shown in Figure 4b. Conversely, if we decide it is a colliding variable, we should not control for it and should use the correlation in Figure 4a. Identifying confounding and colliding variables requires a different approach to causal inference, which is beyond the scope of this paper. Interested readers can refer to key literature on causal diagrams for more information [1,42,65]. The main point here is that correlations could be spurious and, therefore, misleading, so we should be careful when making logical inferences or drawing conclusions based on them. Using causal inference methods would avoid this risk.

7. Discussions

We directly quantified the causal effect of the district heating policy on indoor operative temperature and thermal sensation because this natural experiment allows us to do so. However, it might be inappropriate to infer the indirect causal effect of the indoor operative temperature on the thermal sensation vote. Our assumption is that the district heating policy affects building occupants' thermal sensation votes mainly through the indoor operative temperature, including through consequent changes in relative humidity and clothing insulation. However, it might be possible that the district heating policy affects thermal sensation votes independently of the indoor operative temperature. For example, the perception of subsidized district heating might have a direct and negative psychological impact, such that a higher expectation would lower behavioral adaptations of clothing layers and affect thermal sensations. We suggest further research using another causal inference framework and corresponding algorithms based on causal diagrams [42] to better estimate the indirect causal effect of interest.

It is worth noting that there might not be strict compliance with the policy in regression discontinuity research. Those assigned to the treatment group may not take the treatment, or those assigned to the control group may take the treatment anyway. The fuzzy regression discontinuity method has been developed to deal with the issue [51,52,66]. It requires calculating the different probabilities of the policy assignment in the treatment group and in

the control group at the threshold. In the Chinese thermal comfort dataset, all cities south of the boundary line (the control group) are not provided with district heating, and the probability of district heating is zero. However, there are differences in the extent to which local domestic heating is being used. Three cities north of the boundary line (the treatment group) should be provided with district heating but are not. Due to the small sample size of this non-compliance, their influence on the causal effect measurement is likely trivial. However, further exploration and the application of a fuzzy regression discontinuity design would be valuable.

The opportunity for regression discontinuity because of policy thresholds may seem rare in observational studies. However, it deserves our attention because of the value of assessing interventions in real buildings, and there are many research questions in building science that cannot be answered through experiments to prove causation. Furthermore, we demonstrated that using conventional correlational methods to analyze observational data could lead to spurious results. There are also other natural experiment methods, such as differences in difference [31] and instrument variables [67–69]. We hope that the framework and case study introduced here will encourage scientists and researchers in the building science field to use natural experiments to identify casual effects.

8. Conclusion

We mathematically described causal effects from randomized experiments and generalized them to observational studies based on natural experiments. We demonstrated the application of the regression discontinuity method to a unique thermal comfort field study dataset at China's Huai River, which can be considered a natural experiment of district heating. This approach permitted the estimation of the causal impacts of district heating on both the physical indoor environments and subjective responses of building occupants in winter. We also used conventional correlational analysis and found that the indoor operative temperature could be either positively or negatively correlated with the thermal satisfaction rate, showing an example of Simpson's paradox in building science field studies. This highlights the importance of causal inference frameworks and methods, especially in observational studies. The causal inference framework can be applied in other domains of building science where observational data is collected, and correlational methods are often used, including indoor air quality field studies, post-occupancy evaluation, and building energy audits. This work also has practical implications for sustainable building design and operation that should rely on actionable causal knowledge.

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